

Question number	Answer	Additional guidance	Mark
3(a)(i)	A calculation showing the following steps: • calculation of radius of lesion A (1) • calculation of volume of A and answer given to two decimal places (1)	$A = (1.97 \div 2) = 0.985 / 0.99 / 0.98$ answer to 2 dp between 3.94 and 4.06 correct answer for calculation using their pi value not to 2 dp = 1 mark 4.0 = 1 mark correct answer scores full marks	(2)

These are the most common responses but please always use the mark scheme

1 mark	2 marks
0.985	answer to 2 dp between 3.94 and 4.06
0.99	
0.98	
answer between 3.940457173 to 4.06438 that is not given to 2 dp	

Question number	Answer	Additional guidance	Mark
3(a)(ii)	A calculation showing the following steps: • measurement of X - Y (1) • calculation of actual diameter given to two significant figures (1)	0.55cm (accept 0.53 – 0.60cm) (accept 5.3-6.0mm) Ignore units $(5.5 \div 12 \times 1.97) = 0.90 \text{ (cm)}$ accept answer to 2 sig fig in range 0.87 to 0.99 If within range but not to 2 sig figs = 1 mark eg 0.985 (cm) correct answer scores full marks	(2)

These are the most common responses but please always use the mark scheme

1 mark	2 marks
0.53 to 0.6 (or 0.60)	2 sig fig in range 0.87 to 0.99
5.3 to 6.0 or (6)	
in range 0.87 to 0.99 not to 2 sig fig e.g. 0.985	

Question number	Answer	Additional guidance	Mark
3(b)	An answer that includes the following point: • to enhance the image (1)	to observe the tumour cells to identify where the tumour / tumour cells are to give a clearer image to make it easier to identify / find the tumour cells to see structure in brain to make structure visible to increase contrast / add colour (between tissues) accept lesions for tumour	(1)

Question number	Answer	Additional guidance	Mark
3(c)	An answer that includes two of the following points: • loss of myelin (in motor neurones) (1) • {damage to / lesions in} cerebellum (1) • (electrical) impulses are slowed down / {fewer / less} impulses reach (leg) muscles (1)	ignore signals accept {destruction of / damage to} myelin accept ADEM found in cerebellum accept reduced saltatory conduction accept fewer muscles are stimulated	(2)

